

Quiz 3: Conditional Expectation, Entropy, Stochastic Processes

Homework for lectures 5 and 6. Notation as in the lectures. Twelve problems in three parts, four per part, numbered 1–12.

Part A: Conditional Expectation

Notation as in the Conditional Expectation lecture. For random variables X, Y on $(\Omega, \mathcal{F}, \mathbb{P})$, $\mathbb{E}[X \mid Y = y]$ is the average of X over the row $\{Y = y\}$, and $\mathbb{E}[X \mid Y]$ is the random variable that takes the value $\mathbb{E}[X \mid Y = y]$ on the event $\{Y = y\}$. The results used below are the law of total expectation, the tower property, “taking out what is known”, and the restart property.

1. A random defect rate. A supplier’s defect rate is unknown. Model it as a random variable Θ with the uniform distribution on $[0, \frac{1}{5}]$; its pdf is $f_{\Theta}(\theta) = 5$ for $0 \leq \theta \leq \frac{1}{5}$. A sample of 10 items is inspected. Let D be the number of defective items in the sample. Given $\Theta = \theta$, D is binomial with parameters 10 and θ . Take the following three facts about the binomial distribution as given:

$$\begin{aligned}\mathbb{E}[D \mid \Theta = \theta] &= 10\theta, \\ \mathbb{E}[D^2 \mid \Theta = \theta] &= 10\theta(1 - \theta) + 100\theta^2, \\ \mathbb{P}(D = 0 \mid \Theta = \theta) &= (1 - \theta)^{10}.\end{aligned}$$

- (a) Compute $\mathbb{E}[D]$.
 - (b) Compute $\mathbb{E}[D^2]$ and $\text{Var}(D)$.
 - (c) A binomial random variable with parameters 10 and 0.1 has variance $10 \cdot 0.1 \cdot 0.9 = 0.9$. State whether $\text{Var}(D)$ is larger or smaller than 0.9. Give the reason in one sentence.
 - (d) Compute $\mathbb{P}(D = 0)$ by integrating $\mathbb{P}(D = 0 \mid \Theta = \theta)$ against the pdf of Θ . Compare the result with 0.9^{10} .
2. Waiting for a pattern. Toss a fair coin repeatedly. Fix a pattern of three letters, for example THH. Let T be the number of the toss on which the pattern is first completed. Example: for the tosses H, T, H, H, the pattern THH is completed on toss 4, so $T = 4$. The aim is $\mathbb{E}[T]$ for each of the three patterns THH, THT and TTT.

The method is the one used for HH in the lecture (Application 2). At every moment the game is in one of three states, which record how much of the pattern the most recent tosses have completed. For THH and for THT the states are: state 0, no progress; state 1, the last toss is T; state 2, the last two tosses are TH. For TTT the states are: state 0, no progress; state 1, the last toss is T; state 2, the last two tosses are TT. Let m_j be the expected number of further tosses until the pattern is completed, starting from state j . Then $\mathbb{E}[T] = m_0$.

- (a) For the pattern THH, condition on the next toss to write one equation for each of m_0, m_1, m_2 . Solve the three equations.
- (b) Do the same for THT.

- (c) Do the same for TTT.
- (d) State which of the three patterns has the smallest $\mathbb{E}[T]$.
- (e) The three patterns have the same length. Explain in at most three sentences why their expected waiting times differ. Hint: when an attempt at the pattern fails, how much of the pattern is still in place?
3. Sum and difference of two dice. Two fair dice are rolled. Y is the first die and Z the second. Let $X = Y + Z$ and $D = Y - Z$. The two-dice table has 36 cells (y, z) with $y, z \in \{1, \dots, 6\}$, each of probability $\frac{1}{36}$. Recall $\mathbb{E}[X] = 7$ and $\mathbb{E}[D] = 0$.
- (a) Fix $x \in \{2, \dots, 12\}$. The event $\{X = x\}$ is the set of cells (y, z) with $y + z = x$. Show that if the cell (y, z) belongs to $\{X = x\}$ then so does the cell (z, y) , and that D takes opposite values on these two cells. Deduce that $\mathbb{E}[D \mid X = x] = 0$ for every x , so that $\mathbb{E}[D \mid X] = 0$.
- (b) Fix $d \in \{-5, \dots, 5\}$. The event $\{D = d\}$ is the set of cells (y, z) with $y - z = d$. List these cells and the value of X on each. Show that $\mathbb{E}[X \mid D = d] = 7$ for every d , so that $\mathbb{E}[X \mid D] = 7$.
- (c) Use (b), “taking out what is known” and the tower property to show that $\mathbb{E}[XD] = 0$. Deduce that $\text{Cov}(X, D) = 0$.
- (d) Show that X and D are not independent. Use either of the following two methods. Method 1: show that X and D always have the same parity, then compare $\mathbb{P}(D \text{ odd}, X \text{ even})$ with $\mathbb{P}(D \text{ odd})\mathbb{P}(X \text{ even})$. Method 2: compute $\mathbb{P}(D = 5, X = 7)$ and compare it with $\mathbb{P}(D = 5)\mathbb{P}(X = 7)$.
- (e) The lecture states (Property 3, “What Independence Does Not Say”) that $\mathbb{E}[X \mid D] = \mathbb{E}[X]$ does not imply $X \perp\!\!\!\perp D$. Explain in at most three sentences how (b) and (d) together give an example of this.
4. A random number of rolls. Roll a fair die repeatedly. Let $X_1, X_2, \dots$ be the rolls: independent, each uniform on $\{1, \dots, 6\}$, with $\mu = \mathbb{E}[X_i] = 3.5$. Let N be the number of the first roll that shows 5 or 6. Example: the rolls 2, 4, 1, 6, $\dots$ give $N = 4$. Since each roll shows 5 or 6 with probability $\frac{1}{3}$, N is geometric with success probability $\frac{1}{3}$, and $\mathbb{E}[N] = 3$. The value of N is determined by the rolls, so N is not independent of the X_i , and the random-sums theorem of the lecture does not apply. Define

$$\begin{aligned} S &= X_1 + \dots + X_N && \text{(the stopping roll included),} \\ S' &= X_1 + \dots + X_{N-1} && \text{(the stopping roll excluded; } S' = 0 \text{ if } N = 1). \end{aligned}$$

- (a) Compute $\mathbb{E}[S]$ by conditioning on the first roll X_1 . Hint: on the event $\{X_1 = k\}$ with $k \leq 4$, the rolls $X_2, X_3, \dots$ are again a sequence of fair rolls, so $X_2 + \dots + X_N$ has the same distribution as S (restart property).
- (b) Compute $\mu \mathbb{E}[N]$. State whether it equals $\mathbb{E}[S]$.
- (c) Compute $\mathbb{E}[S']$ by the method of (a).
- (d) Compute $\mu \mathbb{E}[N - 1]$. State whether it equals $\mathbb{E}[S']$.
- (e) Compute $\mathbb{E}[X_N]$, the mean of the stopping roll. Then explain in at most three sentences why the product “ μ times the expected number of terms” gives the correct value in (b) but not in (d).

Part B: Entropy

Logarithms are base 2, the unit is the bit, and $0 \log 0 = 0$. Notation as in the Entropy lecture. p_X is the pmf of X . $H(X)$ is the entropy of X , and $H(p)$ the entropy of a pmf p ; $h(p) =$

$p \log \frac{1}{p} + (1-p) \log \frac{1}{1-p}$ is the binary entropy function. For a prefix-free code c with codeword lengths $\ell_c(x)$, $L(c) = \sum_x p_X(x) \ell_c(x)$ is its average length, and L^* is the smallest average length over all prefix-free codes. $H(P, Q) = \sum_x P(x) \log \frac{1}{Q(x)}$ is the cross-entropy of the truth P with the code or model Q , and $D(P \| Q) = H(P, Q) - H(P)$ is the relative entropy. $H(X | Y)$ is conditional entropy and $I(X; Y) = H(X) - H(X | Y)$ is mutual information.

5. Codes for four symbols. A prefix-free code for four symbols has a length profile $(\ell_1, \ell_2, \ell_3, \ell_4)$, the lengths of its four codewords, and its Kraft sum is $\sum_{i=1}^4 2^{-\ell_i}$. The rover's code 0, 10, 110, 111 has profile $(1, 2, 3, 3)$. The fixed-length code 00, 01, 10, 11 has profile $(2, 2, 2, 2)$.

- Let $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$. Using Kraft's inequality and Gibbs' lemma, show that every prefix-free code for four symbols has $L(c) \geq 1.75$ under p . Deduce that the rover's code is optimal for p .
- Show that the only length profiles for four symbols with Kraft sum exactly 1 are $(2, 2, 2, 2)$ and $(1, 2, 3, 3)$, up to the order of the lengths. Hint: order the lengths $\ell_1 \leq \ell_2 \leq \ell_3 \leq \ell_4$ and treat the cases $\ell_1 \geq 2$ and $\ell_1 = 1$ separately.
- Take as given that an optimal code has Kraft sum exactly 1 (if the sum were smaller, some codeword could be shortened). Let p be any pmf on four symbols, with its probabilities written in decreasing order $p_{(1)} \geq p_{(2)} \geq p_{(3)} \geq p_{(4)}$. Show that

$$L^* = \min(2, p_{(1)} + 2p_{(2)} + 3p_{(3)} + 3p_{(4)}).$$

- Apply (c) to $p = (0.45, 0.25, 0.15, 0.15)$. State which code is optimal and the value of L^* . Compute $H(p)$.
 - Apply (c) to $p = (0.3, 0.3, 0.2, 0.2)$. State which code is optimal and the value of L^* . Compute $H(p)$.
 - The source coding theorem states $H(p) \leq L^* < H(p) + 1$. For each of the two pmfs in (d) and (e), write down the interval for L^* that this gives. For each pmf, state whether the average lengths of both candidate codes (fixed-length and rover's) lie inside the interval.
6. Two coins. Coin A shows heads with probability 0.9. Coin B is fair. One of the two coins is chosen uniformly at random; call the chosen coin $C \in \{A, B\}$. The chosen coin is tossed once; call the result $X \in \{H, T\}$.

- Compute the pmf of X and the entropy $H(X)$.
 - Compute $H(X | C = A)$, $H(X | C = B)$ and $H(X | C)$.
 - State which is larger, $H(X)$ or $H(X | C)$. The inequality between them is an instance of $h(\frac{1}{2} \cdot 0.9 + \frac{1}{2} \cdot 0.5) \geq \frac{1}{2}h(0.9) + \frac{1}{2}h(0.5)$. Name the property of h , listed on the lecture's slide "Binary Entropy Function", that this inequality expresses.
 - Compute $I(X; C) = H(X) - H(X | C)$.
 - Compute $\mathbb{P}(C = A | X = H)$ and $\mathbb{P}(C = A | X = T)$. Use them to compute $H(C | X)$. Check that $H(C) - H(C | X)$ equals the value of $I(X; C)$ found in (d).
7. Two missions. The rover has two possible missions, $M \in \{1, 2\}$, each with probability $\frac{1}{2}$. On mission m the instructions $X \in \{\text{up}, \text{down}, \text{left}, \text{right}\}$ have pmf P_m , where

$$P_1 = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}), \quad P_2 = (\frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}) \quad \text{on } (\text{up}, \text{down}, \text{left}, \text{right}).$$

Note that $H(P_1) = H(P_2) = 1.75$ bits. One code, built for a pmf Q , must be installed before the mission is known. On mission m it costs $H(P_m, Q)$ bits per instruction, so its expected cost is $\frac{1}{2}H(P_1, Q) + \frac{1}{2}H(P_2, Q)$.

- (a) Show that the expected cost equals $H(\bar{P}, Q)$, where $\bar{P} = \frac{1}{2}(P_1 + P_2)$. Compute $\bar{P}$.
 - (b) Find the pmf Q that minimises the expected cost. Compute the minimal expected cost.
 - (c) Compute the expected cost for $Q = P_1$ (the old code, built for mission 1, is kept) and for $Q = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ (the fixed-length code). Compare both with the minimal cost from (b) and with $\frac{1}{2}H(P_1) + \frac{1}{2}H(P_2) = 1.75$, the cost if the mission were known before the code was chosen.
 - (d) On mission m the code of (b) wastes $D(P_m \parallel \bar{P})$ bits per instruction. Show that the expected waste $\frac{1}{2}D(P_1 \parallel \bar{P}) + \frac{1}{2}D(P_2 \parallel \bar{P})$ equals $I(M; X)$, the mutual information between the mission M and one instruction X . Compute it.
 - (e) Codeword lengths must be integers, so the code of (b) cannot be installed as it stands. Problem 5(b) showed that the only length profiles for four symbols with Kraft sum 1 are $(2, 2, 2, 2)$ and $(1, 2, 3, 3)$. Among codes that assign one codeword to each instruction, find the one with the smallest expected cost, and give that cost.
 - (f) Name the method from the lecture (slide “Closing the Gap: Code Blocks of Symbols”) by which the cost per instruction can be brought closer to the minimal cost of (b).
8. A Markov text source. A binary text is modelled as a Markov chain $X_1, X_2, \dots$ on $S = \{0, 1\}$ with transition matrix

$$Q = \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix},$$

where row x is the conditional pmf of X_{i+1} given $X_i = x$. The chain is started from its stationary pmf π , so that every X_i has pmf π . Recall from the lecture: the entropy rate is $H(X_{i+1} \mid X_i) = \sum_{x \in S} \pi(x) H(Q(x, \cdot))$, the monographic entropy is $H(\pi)$, and the perplexity of a model that needs b bits per symbol is 2^b .

- (a) Find the stationary pmf π .
- (b) Compute the entropy rate $H(X_{i+1} \mid X_i)$.
- (c) Compute the monographic entropy $H(\pi)$ and the difference $H(\pi) - H(X_{i+1} \mid X_i)$. State which quantity from the lecture this difference is.
- (d) Compute $H(X_1, X_2, X_3)$ by the chain rule, using the Markov property.
- (e) Compute $H(X_1, X_2, X_3)$ again by listing the probabilities of the eight possible strings $x_1 x_2 x_3$ and applying the definition of entropy.
- (f) Compute the perplexity of the Markov model, of the monographic model and of the uniform model.
- (g) A compressor built on the monographic model uses a codeword of length $\log \frac{1}{\pi(x)}$ for the symbol x at every position. Compute its cost in bits per symbol on this text. Compute its waste, that is, the cost minus the entropy rate.

Part C: Stochastic Processes

Notation as in the Stochastic Processes lecture. B is a standard Brownian motion, so B_t has the $N(0, t)$ distribution. $\mathbb{P}_i$ and $\mathbb{E}_i$ refer to a chain started at state i . In the gambler’s ruin, p is the probability of winning a round, $q = 1 - p$, and $r = q/p$. $\mathbf{P} = (p(i, j))$ is the transition matrix; row i is the conditional pmf of the next state given the current state i . π is a stationary distribution, $\pi \mathbf{P} = \pi$. For a target set A , $h_i = \mathbb{P}_i(\text{reach } A)$ is the hitting probability and $k_i = \mathbb{E}_i[\text{number of steps until } A \text{ is reached}]$ is the expected hitting time; both satisfy the first-step systems of the lecture. Φ is the cdf of $N(0, 1)$; the values needed below are

$$\Phi(0.100) = 0.540, \quad \Phi(0.500) = 0.691, \quad \Phi(0.674) = 0.750, \quad \Phi(1.000) = 0.841.$$

9. Hitting times of Brownian motion. The reflection principle states that for $a > 0$ and $t > 0$,

$$\mathbb{P}(\max_{s \leq t} B_s \geq a) = 2\mathbb{P}(B_t \geq a).$$

For $a > 0$ let $T_a = \min\{s \geq 0 : B_s = a\}$ be the first time B reaches the level a . The event $\{T_a \leq t\}$ is the same as the event $\{\max_{s \leq t} B_s \geq a\}$.

- Compute $\mathbb{P}(\max_{s \leq 4} B_s \geq 2)$.
 - Write $\mathbb{P}(T_1 \leq t)$ in terms of Φ . Evaluate it at $t = 1$, $t = 4$ and $t = 100$.
 - The median of T_1 is the value t with $\mathbb{P}(T_1 \leq t) = \frac{1}{2}$. Find it.
 - The scaling property of the lecture states that $(\frac{1}{a}B_{a^2t})_{t \geq 0}$ is again a Brownian motion. Use it to show that T_a has the same distribution as a^2T_1 . Give the median of T_a .
 - The lecture showed that $\mathbb{E}[T_1] = \infty$. Explain in at most three sentences how an infinite mean is consistent with the median found in (c). You may use the approximation $\mathbb{P}(T_1 > t) \approx \sqrt{2/(\pi t)}$ for large t , from the lecture.
10. Bold play. A gambler wins each round with probability $p = 0.4$ and loses it with probability $q = 0.6$, so $r = q/p = 1.5$. The fortune starts at $\mathcal{L}3$ and the target is $\mathcal{L}10$; the game ends when the fortune reaches $\mathcal{L}0$ or $\mathcal{L}10$. Two strategies are compared. Timid play stakes $\mathcal{L}1$ per round. From the lecture, with $N = 10$ and starting fortune i ,

$$h_i = \frac{r^i - 1}{r^N - 1}, \quad \mathbb{E}_i[T] = \frac{i}{q - p} - \frac{N}{q - p} \cdot \frac{1 - r^i}{1 - r^N},$$

where T is the number of rounds until the game ends. Bold play stakes, from fortune i , the amount $\min(i, 10 - i)$: as much as possible without overshooting the target. A win adds the stake to the fortune and a loss removes it. So the fortune moves from 3 to 6 (win) or 0 (loss); from 6 to 10 or 2; from 2 to 4 or 0; from 4 to 8 or 0; from 8 to 10 or 6.

- Timid play: compute h_3 and $\mathbb{E}_3[T]$ from the two formulas above.
 - Bold play: draw the chain on the states $\{0, 2, 3, 4, 6, 8, 10\}$ with the transition probabilities. Write the first-step system for $h_i = \mathbb{P}_i(\text{reach } 10)$.
 - Solve the system of (b) for h_3 . Compare with the timid h_3 of (a).
 - Bold play: write the first-step system for $k_i = \mathbb{E}_i[\text{number of bets until the game ends}]$ and solve it for k_3 . Compare with the timid $\mathbb{E}_3[T]$ of (a).
 - Now let the game be fair, $p = q = \frac{1}{2}$. Solve the bold-play system of (b) for h_3 . For timid play the lecture gives $h_3 = \frac{3}{10}$. Explain in at most three sentences why the two values agree.
11. PageRank. A random surfer on n pages moves as follows: with probability $d = 0.85$ it follows a uniformly chosen link out of the current page, and with probability $1 - d = 0.15$ it jumps to a uniformly chosen page. Its transition matrix is

$$\mathbf{P} = d\mathbf{H} + \frac{1 - d}{n}\mathbf{J},$$

where $\mathbf{H}_{ij}$ is $1/(\text{number of links out of } i)$ if page i links to page j and 0 otherwise, and $\mathbf{J}$ is the matrix of ones. The PageRank of a page is its entry in the stationary distribution π of $\mathbf{P}$.

- Three pages with links $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow A$. Write $\mathbf{H}$ and $\mathbf{P} = 0.85\mathbf{H} + 0.05\mathbf{J}$.
- Solve $\pi\mathbf{P} = \pi$, $\pi_A + \pi_B + \pi_C = 1$, and rank the three pages.

- (c) Pages A and B each have a single in-link. Explain in at most two sentences why $\pi_A > \pi_B$.
- (d) Find the stationary distribution of $\mathbf{H}$ itself, that is, without damping ($d = 1$).
- (e) Add a fourth page D with the links $C \rightarrow D$ (so C now links to both A and D) and $D \rightarrow D$, and no other links out of D. Find the stationary distribution without damping ($d = 1$) and with $d = 0.85$ (now $n = 4$, so $\mathbf{P} = 0.85\mathbf{H} + 0.0375\mathbf{J}$).
- (f) Describe in at most three sentences what the page D does to the surfer when $d = 1$, and what the damping changes.
12. Tennis at deuce. From deuce, the server wins each point with probability p and loses it with probability $q = 1 - p$, independently of the other points. The game is won by the first player to lead by two points. The states are D (deuce), I (advantage server), O (advantage receiver), W (game won by the server) and L (game lost by the server). From D the next state is I or O; from I it is W or D; from O it is D or L; W and L are absorbing.
- (a) Write the transition matrix, with the states in the order D, I, O, W, L.
- (b) Write the first-step system for $h_i = \mathbb{P}_i(\text{reach W})$. Solve it and show that $h_D = \frac{p^2}{p^2 + q^2}$.
- (c) Write the first-step system for $k_i = \mathbb{E}_i[\text{number of points until the game ends}]$. Solve it and show that $k_D = \frac{2}{1 - 2pq}$.
- (d) Evaluate h_D and k_D for $p = 0.5, 0.55, 0.6$ and 0.7 . For $p = 0.55$, state h_D as a percentage.
- (e) Let $m \geq 1$. The game ends after exactly $2m$ further points if the first $m - 1$ pairs of points return to deuce and the m -th pair does not. Write h_D as an infinite series by summing over m . Then explain in at most two sentences why the first-step system of (b) gives h_D without summing a series.